

eBEAM LITHOGRAPHY: A Breakthrough For Compound Semiconductor Fabrication



eBeam technology catalyses innovation across industries, from driving efficient EVs to advancing quantum computing. Its integration marks a transformative leap in precision manufacturing.



The author, **VINAYAK RAMACHANDRA ADKOLI**, is BE in industrial production and has served as a lecturer in three different polytechnics for ten years. He is also a freelance writer and cartoonist

Innovation propels electronics forward, meeting the growing demand for faster, smaller, and energy-efficient devices. Compound semiconductors—like gallium arsenide (GaAs), gallium nitride (GaN), and silicon carbide (SiC)—are powering advancements in 5G, electric vehicles, renewable energy, and optoelectronics.

Unlocking their full potential, however, requires evolving manufacturing processes. Electron beam (eBeam) technology, a precision-driven solution, promises to transform compound semiconductor production.

Why compound semiconductors?

Compound semiconductors differ from traditional silicon-based semiconductors

because they handle high frequencies, high power, and optoelectronic applications with exceptional efficiency. These materials are critical for cutting-edge technologies, including:

5G networks. Supporting high-frequency transistors and RF components that are essential for faster data transmission

Electric vehicles. Enabling efficient power devices for inverters, chargers, and other components

Renewable energy. Powering high-efficiency energy conversion systems, such as solar inverters and wind turbines

Optoelectronics. Driving advancements in lasers, LEDs, and